1. Apply the Banker's algorithm to check whether the system is in a safe state:
   Available resources: A=3, B=3, C=2
   Processes: Allocation Max Need

   P0: 0 1 0 7 5 3 ?

   P1: 2 0 0 3 2 2 ?

   P2: 3 0 2 9 0 2 ?

   P3: 2 1 1 2 2 2 ?

   P4: 0 0 2 4 3 3 ?

2. Reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2

   Number of frames: 3

   Calculate:

   Number of page faults using:
   FIFO
   LRU
   Optimal replacement

3. Given the following processes, compute Waiting Time, Turnaround Time, Average Waiting Time, and Average Turnaround Time using FCFS, SJF, RR (time quantum = 2), and Priority Scheduling.

Process	Arrival Time	Burst Time	Priority
P1	0 ms	6 ms	3
P2	1 ms	8 ms	1
P3	2 ms	7 ms	2
P4	3 ms	3 ms	4

4. Describe memory management techniques: Paging, Segmentation, and Virtual Memory.
5. Compare contiguous and non-contiguous memory allocation techniques.
6. What is the difference between monolithic kernel and microkernel architecture?
7. Discuss the impact of context switching on system performance.
8. What are the limitations or assumptions of Peterson's solution?
9. Write a short note on:
   a. Context Switching
   b. Kernel
   c. Race Condition
   d. Swapping
   e. Thrashing
   f. Interrupt
   g. Convoy Effect
   h. Belady's Anomaly
   i. Busy Waiting